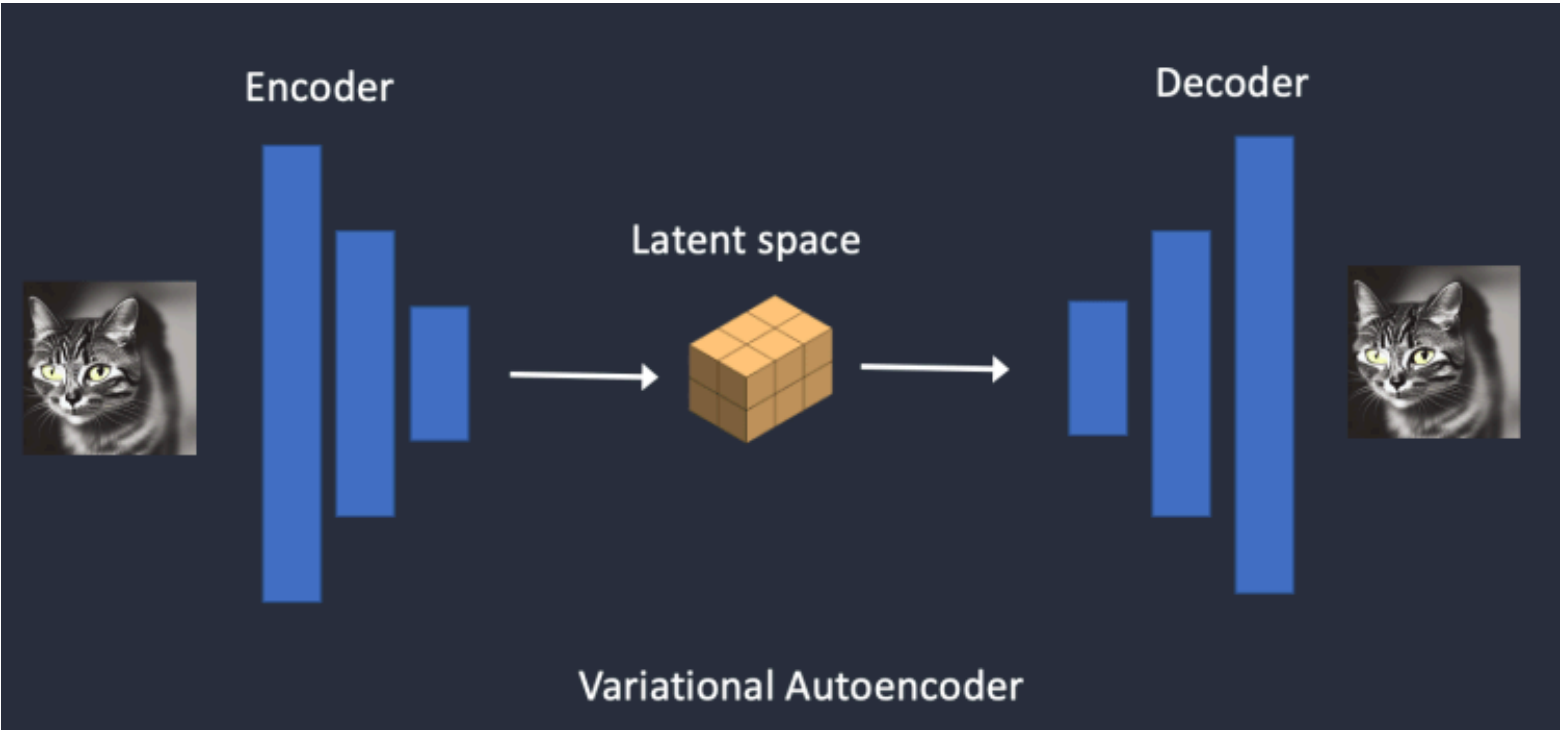
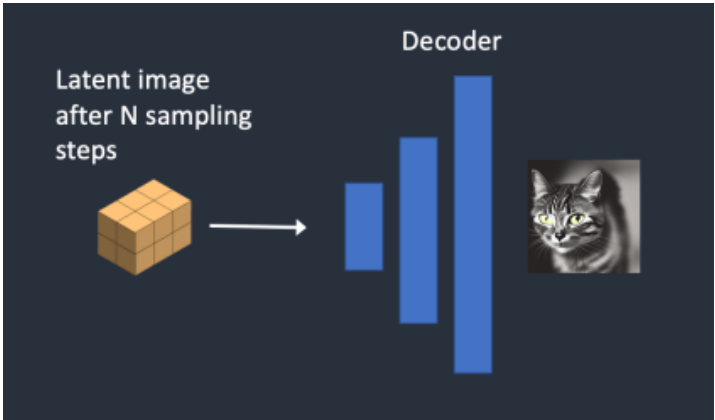
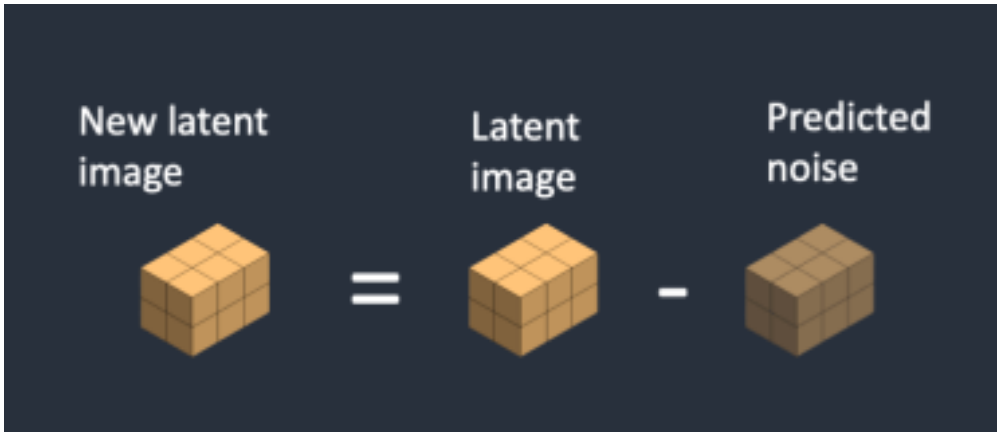
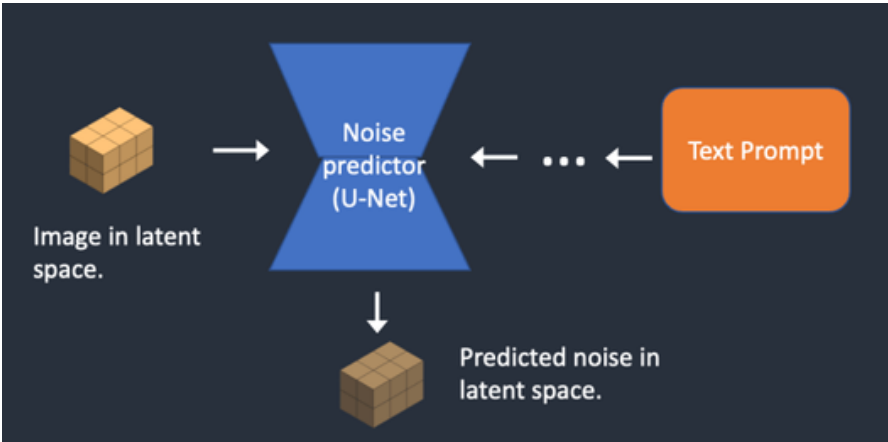
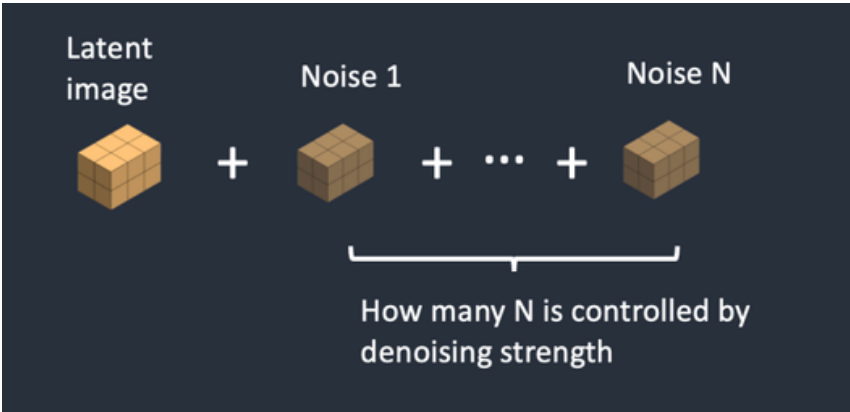


Stable Diffusion

image-to-image

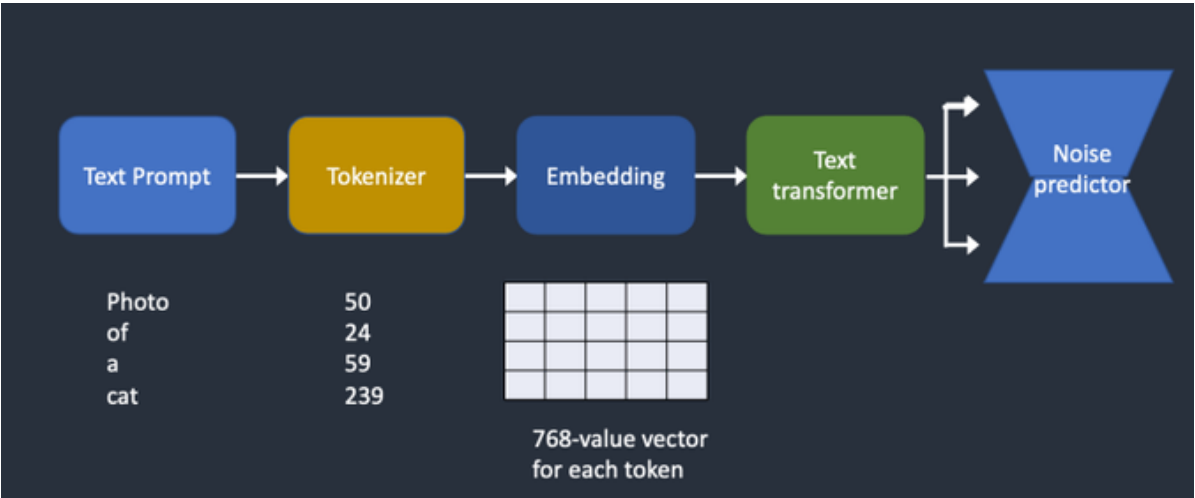


Noise schedule



VQ-VAE

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon_{\theta}(z_t, t, \tau_{\theta}(y)) \right\|_2^2 \right] \quad L_{Autoencoder} = \min_{\epsilon, D} \max_{\psi} \left(\underbrace{L_{rec}(x, D(\epsilon(x)))}_{\text{Reconstruction Loss}} - \underbrace{L_{adv}(D(\epsilon(x))) + \log D_{\psi}(x)}_{\text{Adversarial Objective}} + \underbrace{L_{reg}(x; \epsilon, D)}_{\text{Regularization Loss}} \right)$$



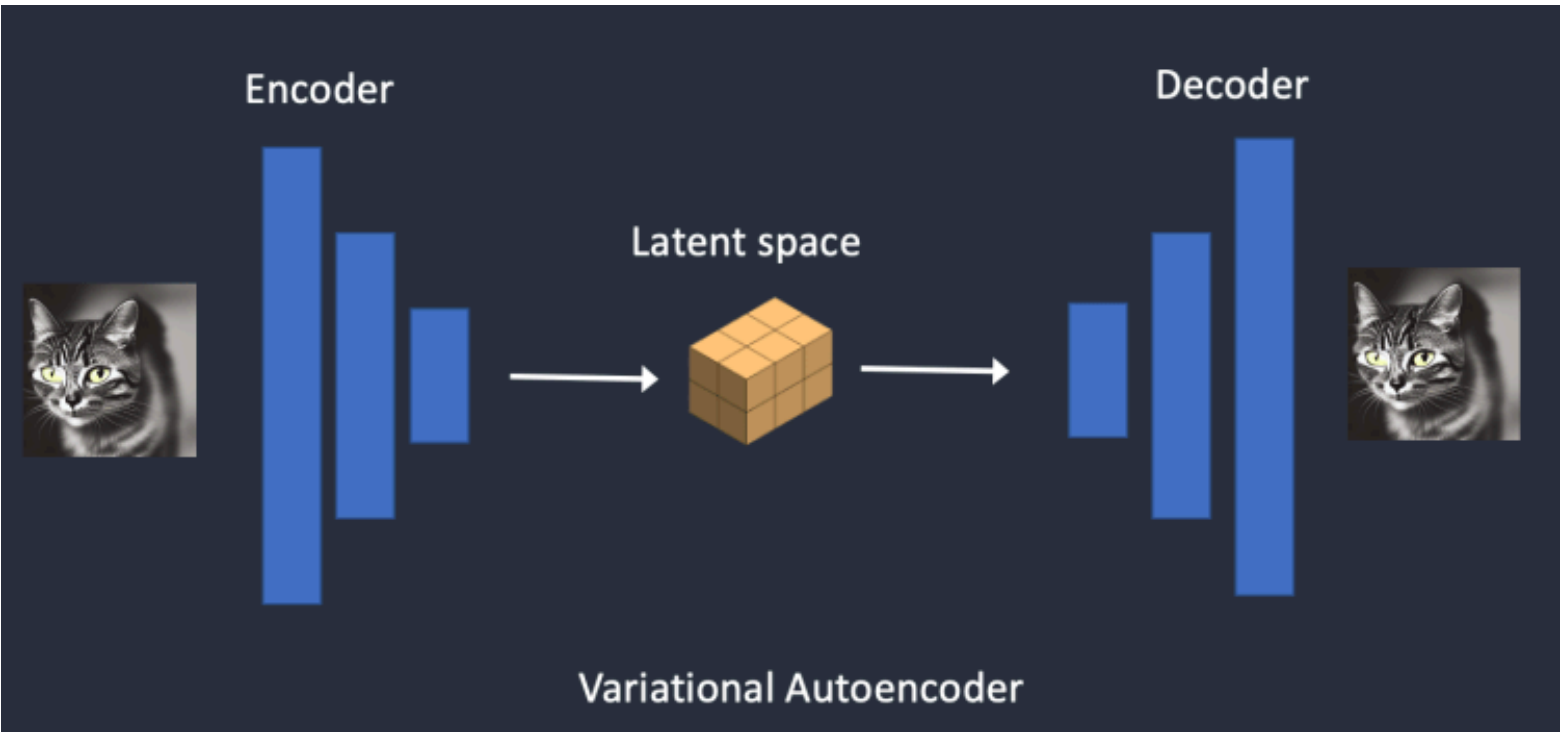
cross-attention

Conditioning

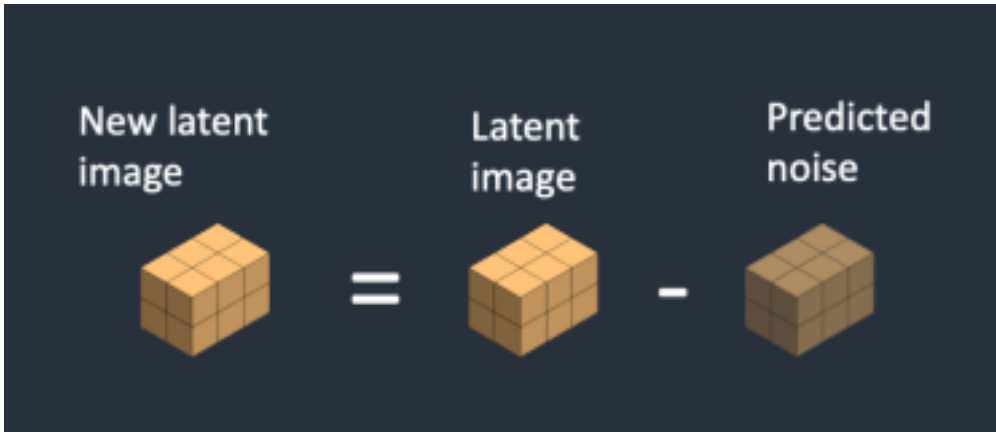
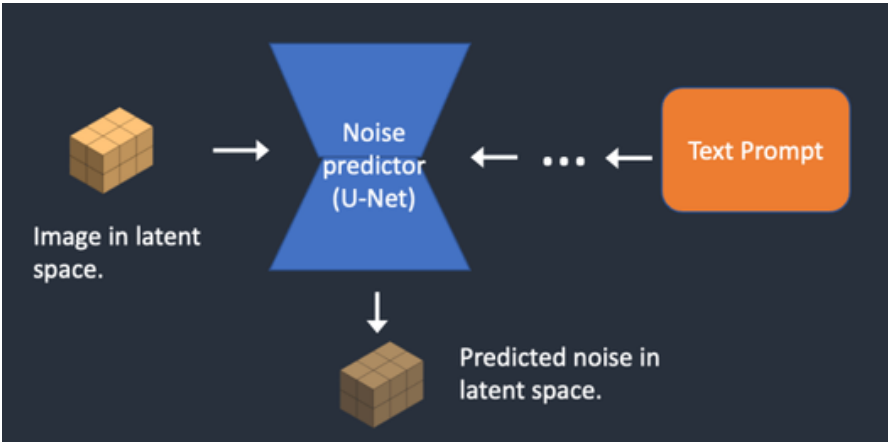
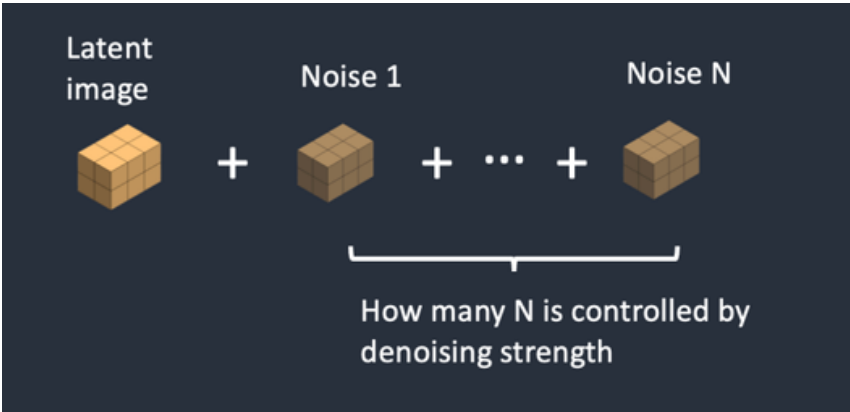
Application : textual inversion

Stable Diffusion(Loss function)

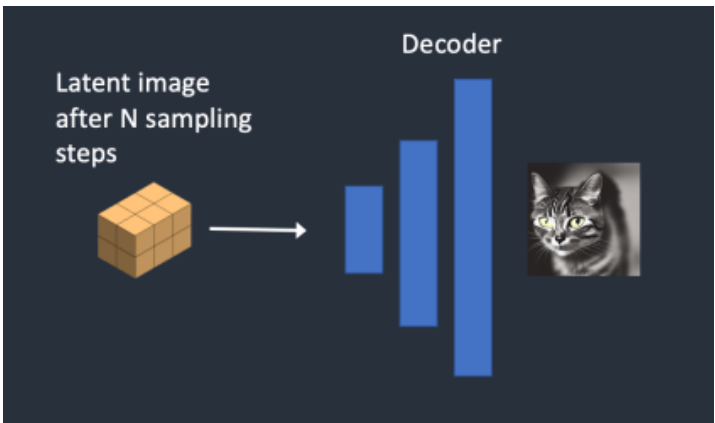
image-to-image



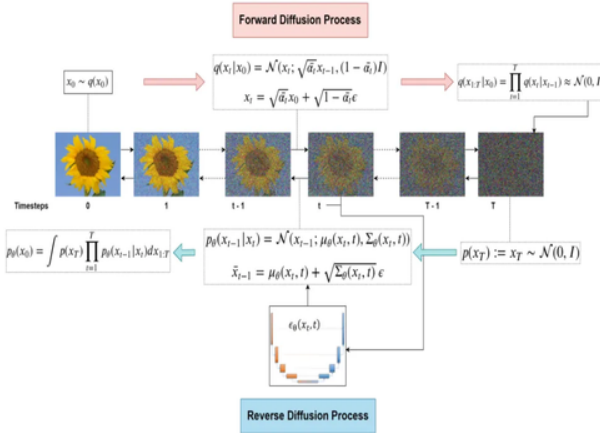
Noise schedule



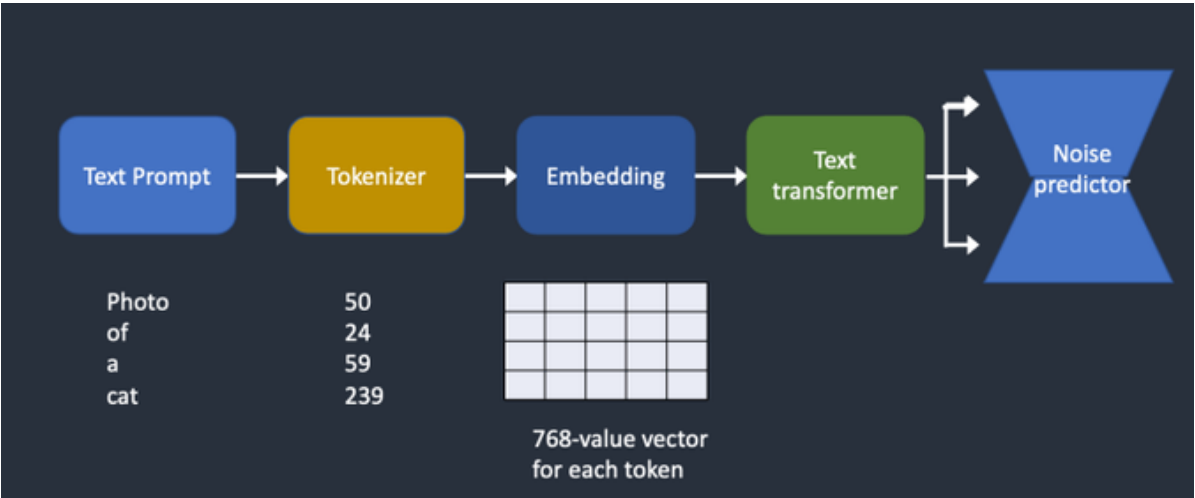
VQ-VAE



$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon_{\theta}(z_t, t, \tau_{\theta}(y)) \right\|_2^2 \right]$$
$$L_{Autoencoder} = \min_{\epsilon, D} \max_{\psi} \left(\underbrace{L_{rec}(x, D(\epsilon(x)))}_{\text{Reconstruction Loss}} - \underbrace{L_{adv}(D(\epsilon(x)))}_{\text{Adversarial Objective}} + \underbrace{\log D_{\psi}(x) + L_{reg}(x; \epsilon, D)}_{\text{Regularization Loss}} \right)$$



cross-attention



Conditioning

Application : textual inversion

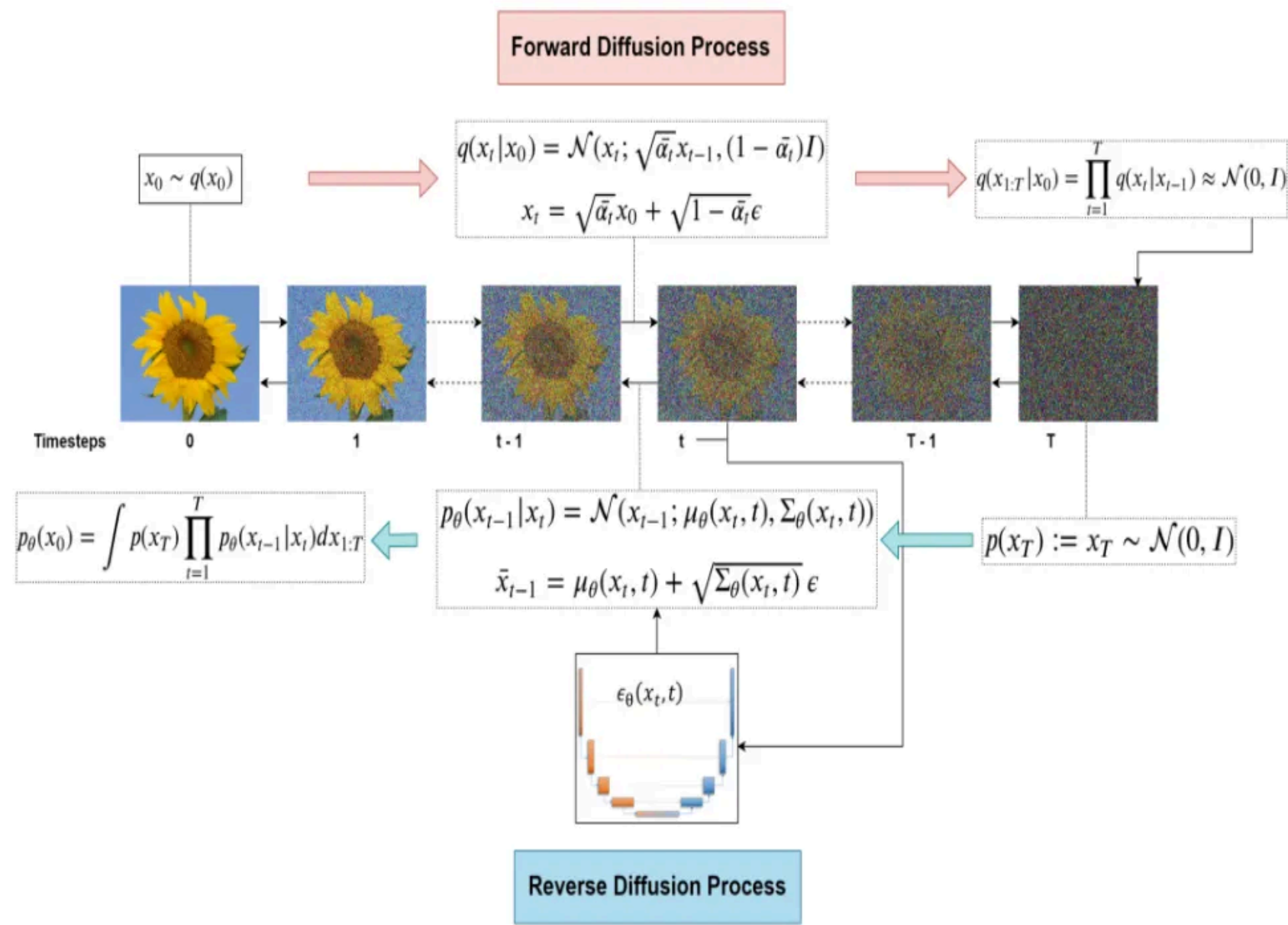
Discriminator(Adversarial loss)

L1/L2 Reconstruction loss + Perceptual Loss + Adversarial Loss

Latent Space Regularization

KL-regularization

VQ-regularization



$$L_{Autoencoder} = \min_{\epsilon, D} \max_{\psi} \left(\underbrace{L_{rec}(x, D(\epsilon(x)))}_{\text{Reconstruction Loss}} - \underbrace{L_{adv}(D(\epsilon(x)))}_{\text{Adversarial Objective}} + \underbrace{\log D_\psi(x) + L_{reg}(x; \epsilon, D)}_{\text{Regularization Loss}} \right)$$

1. L1 & L2 loss:

$$L1 \text{ Loss } (J) = (1/n) * \sum |y_{\text{true}} - y_{\text{pred}}|$$

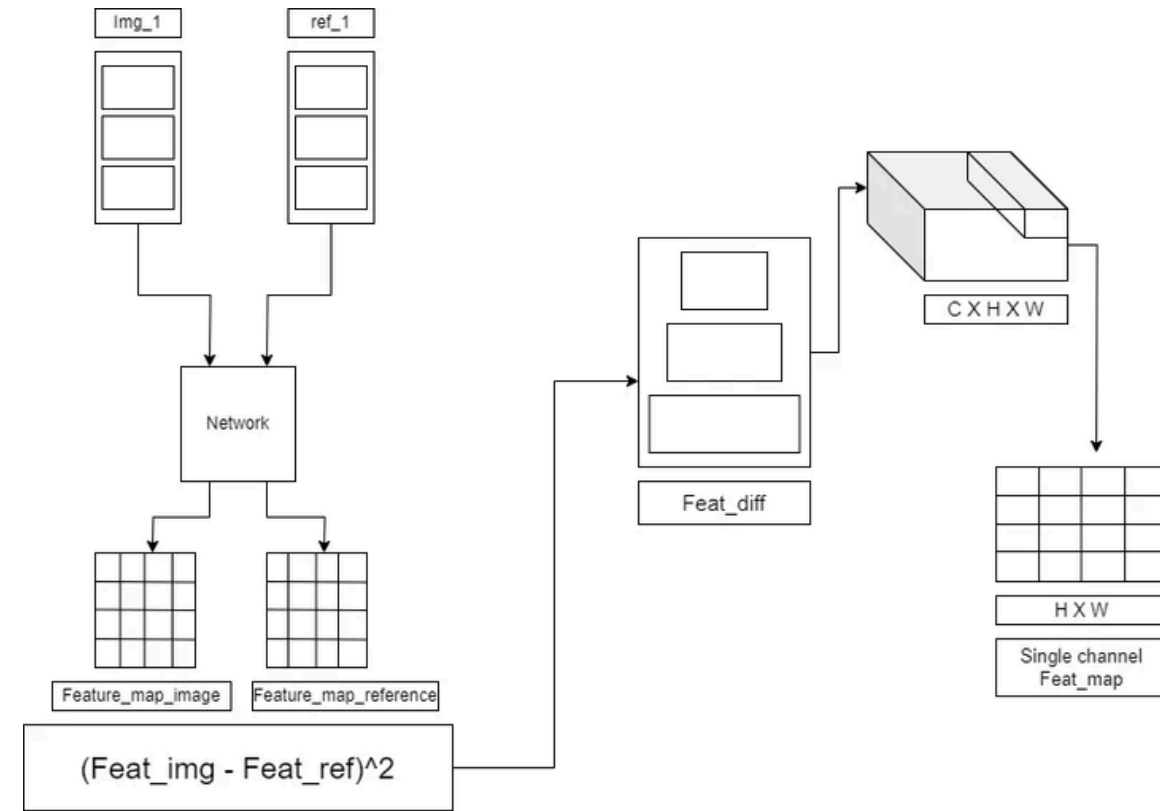
$$MSE \text{ Loss } (J) = (1/n) * \sum (y_{\text{true}} - y_{\text{pred}})^2$$

2. Perceptual Loss:

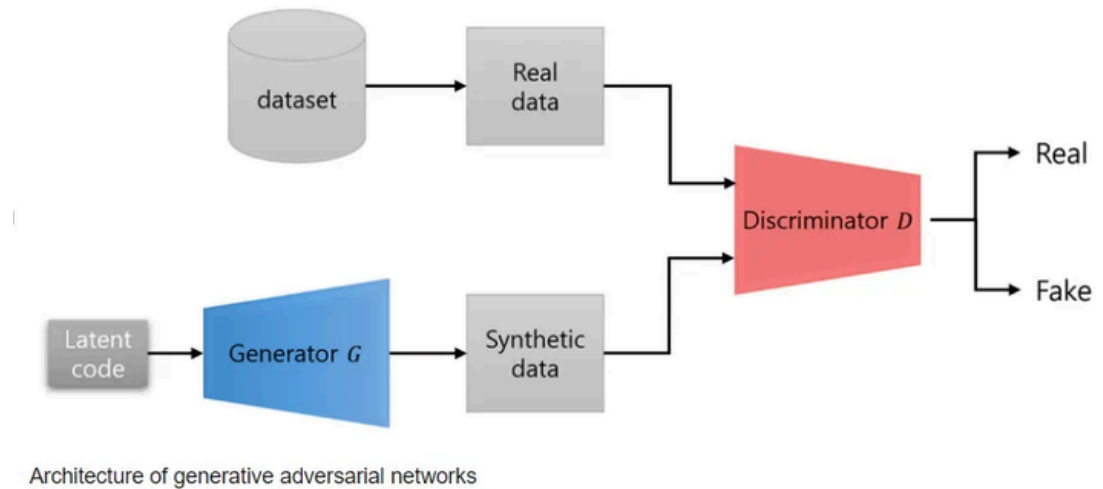
$$\text{Perceptual Loss} = \alpha \times \text{Content Loss} + \beta \times \text{Style Loss}$$

$$d = \frac{1}{H \times W} \sum_{i,j} (\text{Difference}_{i,j})$$

2. Perceptual Loss:



3. Adversarial Loss: generator loss + patch based discriminator loss

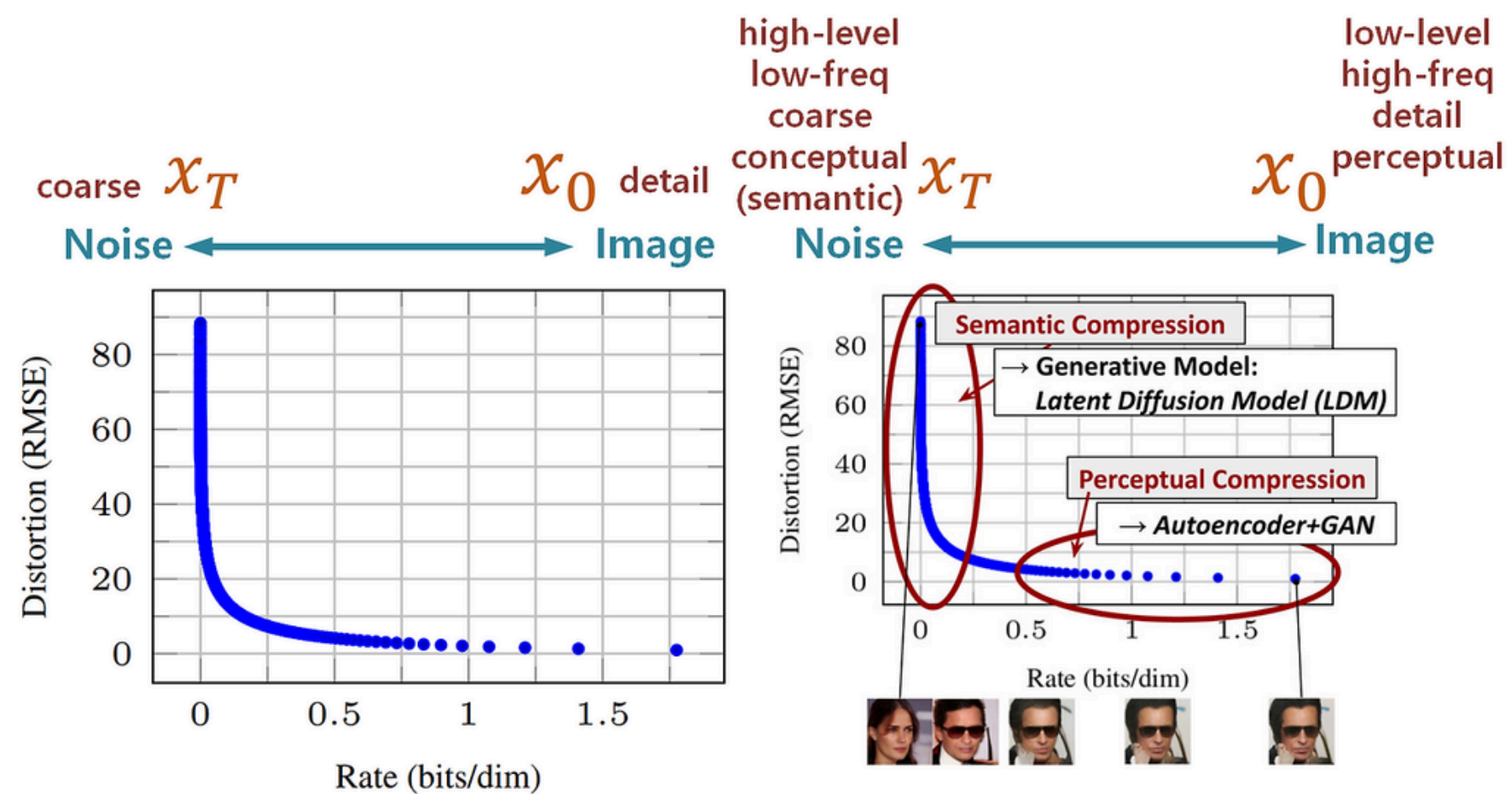


3. Adversarial Loss: generator loss + patch based discriminator loss

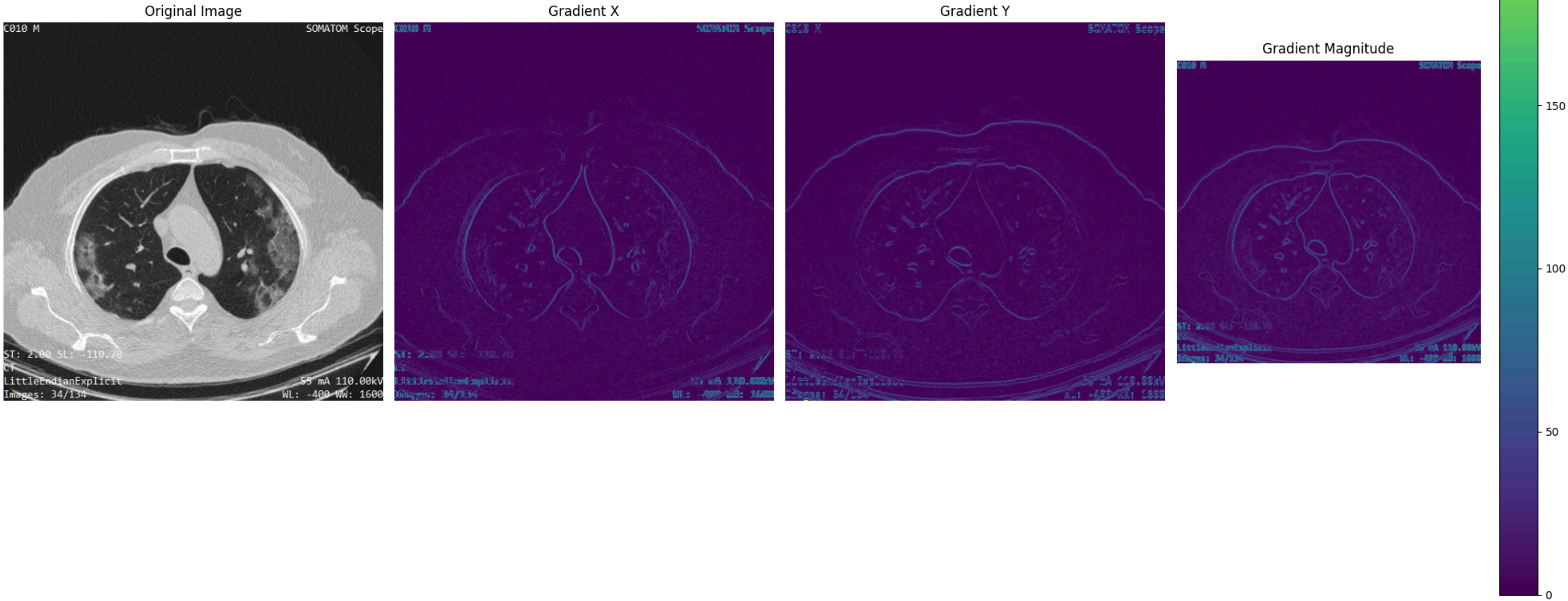
$$\text{Formula for Generator Loss: } L_{adv} = -\log(D(G(z)))$$

$$\text{Formula for Discriminator Loss: } L_{adv} = -[\log(D(x)) + \log(1 - D(G(z)))]$$

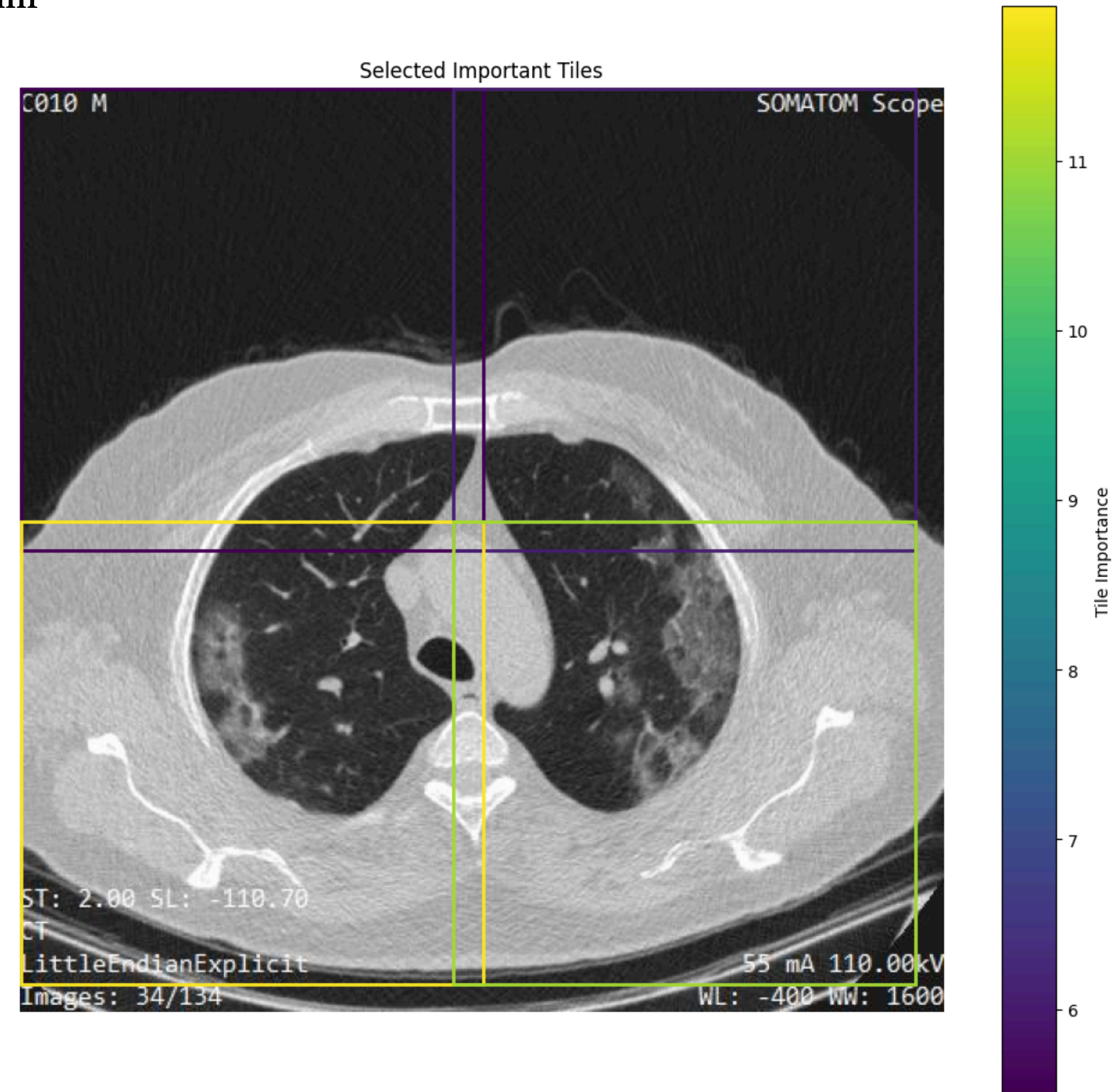
Perceptual equivalent autoencoder and DM



Calculate Graident Magnitude



Why Use Adaptive Tiled Algorithm



=== PSNR 기준 정렬 ===

1. 파일: degraded_NORMAL2-IM-1141-0001.jpeg
업스케일된 파일: upscaled_0_degraded_NORMAL2-IM-1141-0001.jpeg
PSNR: 39.05 dB
SSIM: 0.9480
2. 파일: degraded_IM-0628-0001.jpeg
업스케일된 파일: upscaled_0_degraded_IM-0628-0001.jpeg
PSNR: 38.76 dB
SSIM: 0.9536
3. 파일: degraded_NORMAL2-IM-0690-0001.jpeg
업스케일된 파일: upscaled_0_degraded_NORMAL2-IM-0690-0001.jpeg
PSNR: 38.65 dB
SSIM: 0.9559
4. 파일: degraded_NORMAL2-IM-0831-0001.jpeg
업스케일된 파일: upscaled_0_degraded_NORMAL2-IM-0831-0001.jpeg
PSNR: 38.54 dB
SSIM: 0.9411
5. 파일: degraded_IM-0324-0001.jpeg
업스케일된 파일: upscaled_0_degraded_IM-0324-0001.jpeg
PSNR: 38.51 dB
SSIM: 0.9575

=== SSIM 기준 정렬 ===

1. 파일: degraded_IM-0399-0001.jpeg
업스케일된 파일: upscaled_0_degraded_IM-0399-0001.jpeg
PSNR: 37.86 dB
SSIM: 0.9613
2. 파일: degraded_IM-0370-0001.jpeg
업스케일된 파일: upscaled_0_degraded_IM-0370-0001.jpeg
PSNR: 37.45 dB
SSIM: 0.9583
3. 파일: degraded_NORMAL2-IM-1379-0001.jpeg
업스케일된 파일: upscaled_0_degraded_NORMAL2-IM-1379-0001.jpeg
PSNR: 36.95 dB
SSIM: 0.9579
4. 파일: degraded_NORMAL2-IM-0468-0001.jpeg
업스케일된 파일: upscaled_0_degraded_NORMAL2-IM-0468-0001.jpeg
PSNR: 38.08 dB
SSIM: 0.9575
5. 파일: degraded_IM-0324-0001.jpeg
업스케일된 파일: upscaled_0_degraded_IM-0324-0001.jpeg
PSNR: 38.51 dB
SSIM: 0.9575

PSNR (Peak Signal-to-noise ratio)

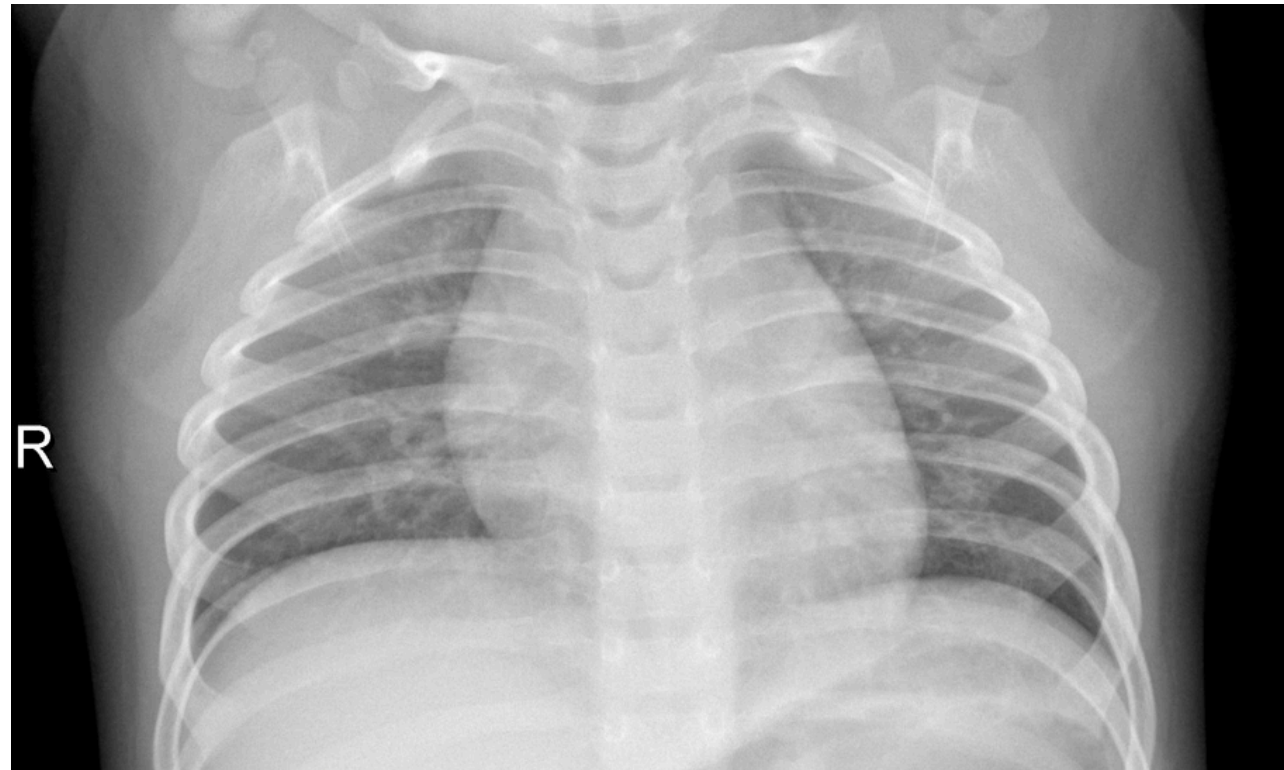
$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad MSE = \frac{\sum_{M,N} [I_1(m,n) - I_2(m,n)]^2}{M * N}$$

SSIM

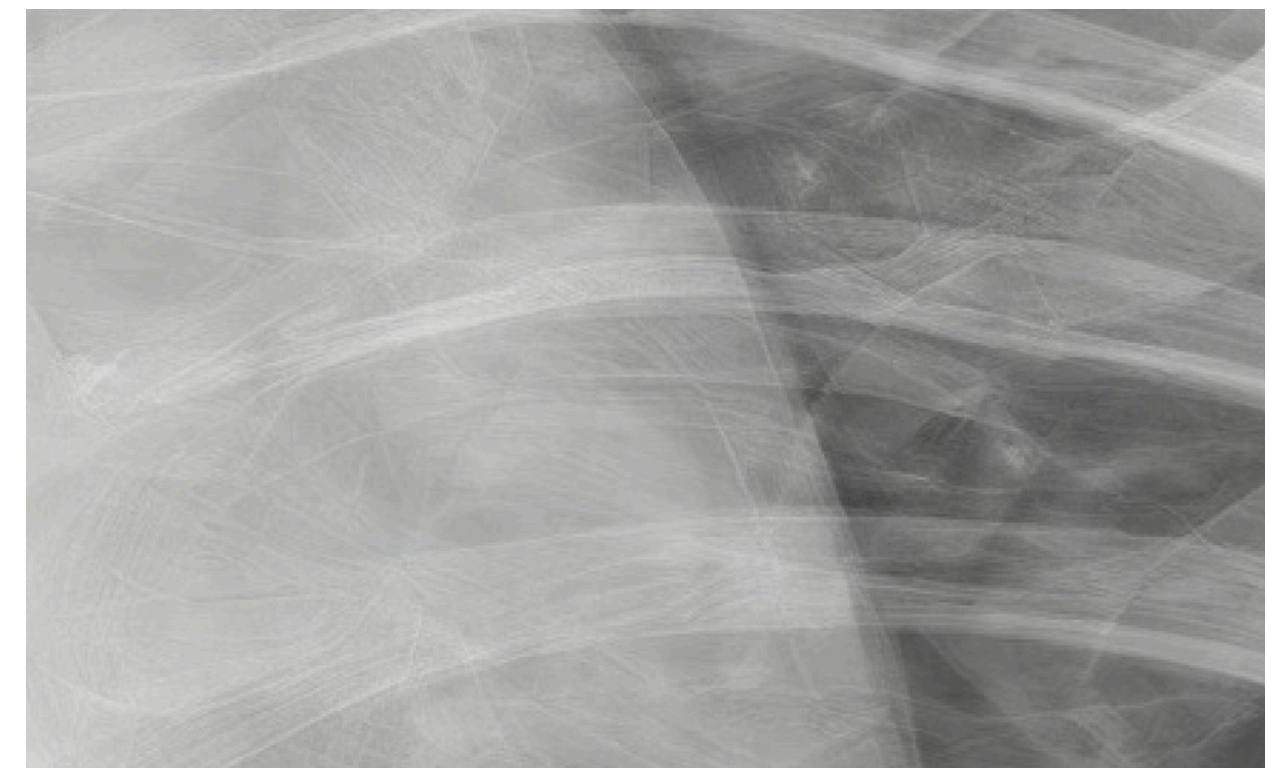
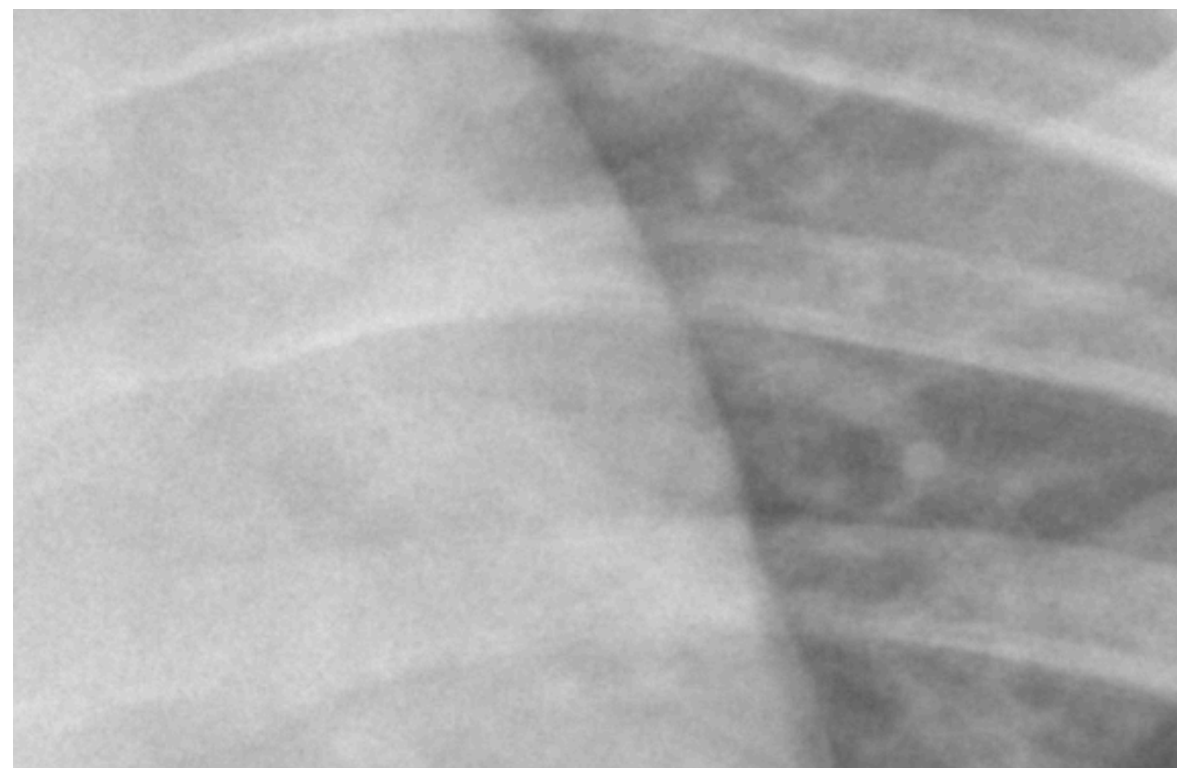
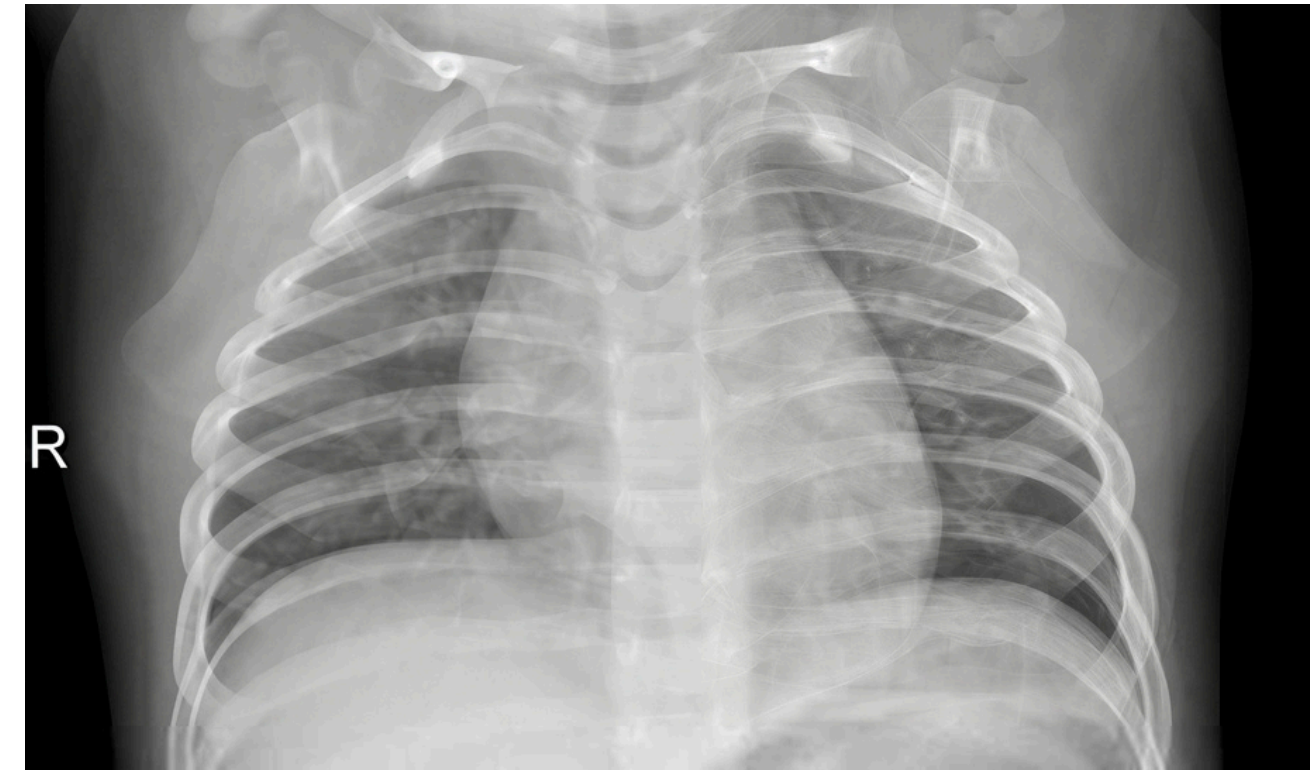
$$SSIM(\mathbf{x}, \mathbf{y}) = [l(\mathbf{x}, \mathbf{y})]^\alpha \cdot [c(\mathbf{x}, \mathbf{y})]^\beta \cdot [s(\mathbf{x}, \mathbf{y})]^\gamma$$

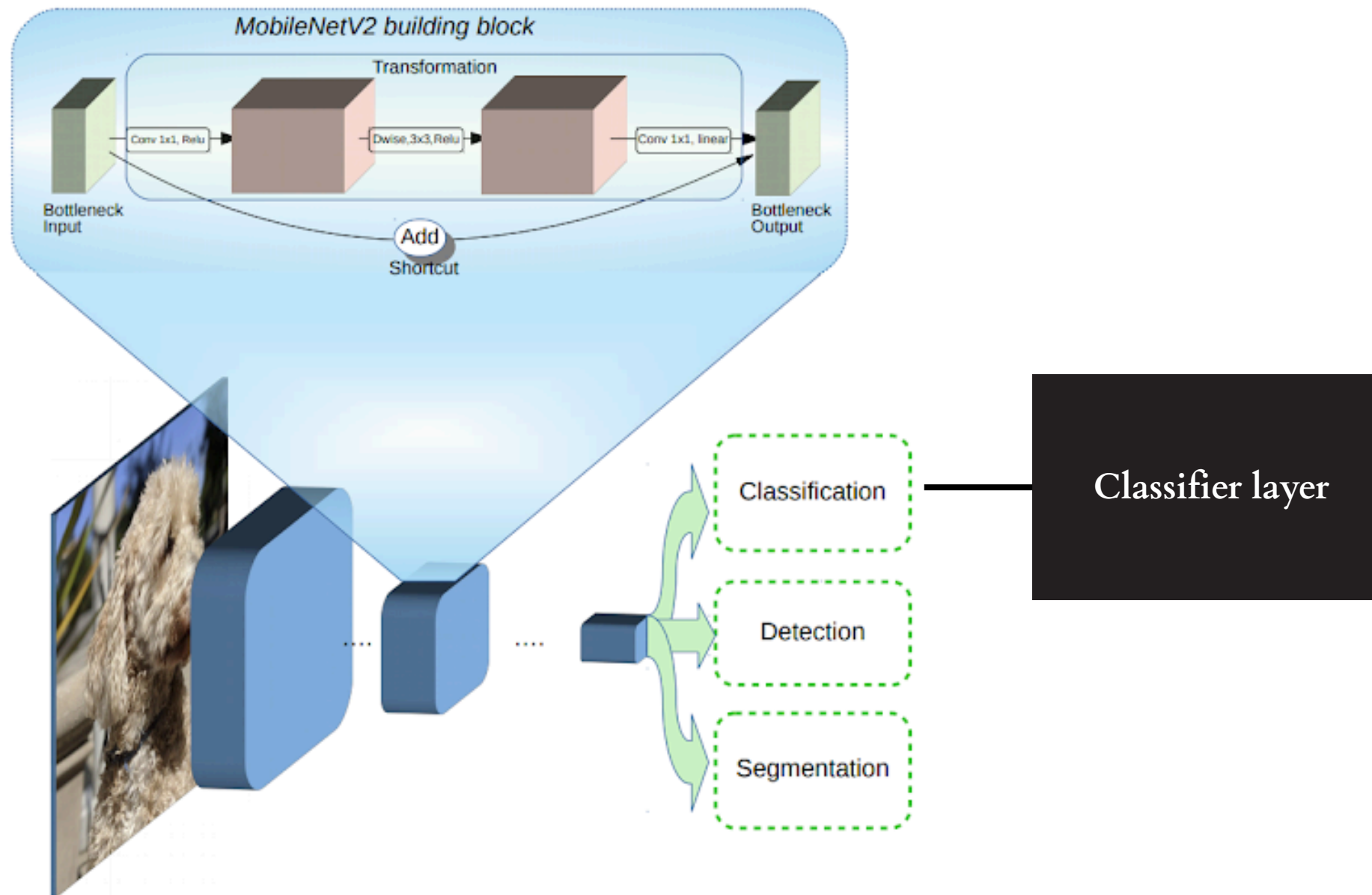
$$l(\mathbf{x}, \mathbf{y}) = \frac{2 \mu_x \mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \quad , \quad c(\mathbf{x}, \mathbf{y}) = \frac{2 \sigma_x \sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \quad , \quad s(\mathbf{x}, \mathbf{y}) = \frac{\sigma_{xy} + C_3}{\sigma_x \sigma_y + C_3}$$
$$\mu_x = \frac{1}{N} \sum_{i=1}^N x_i \quad , \quad \sigma_x = \left(\frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)^2 \right)^{1/2} \quad , \quad \sigma_{xy} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)$$

Ground-Truth Image(1824*1090)

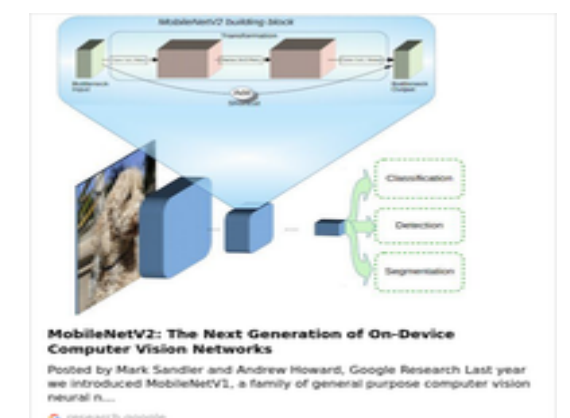
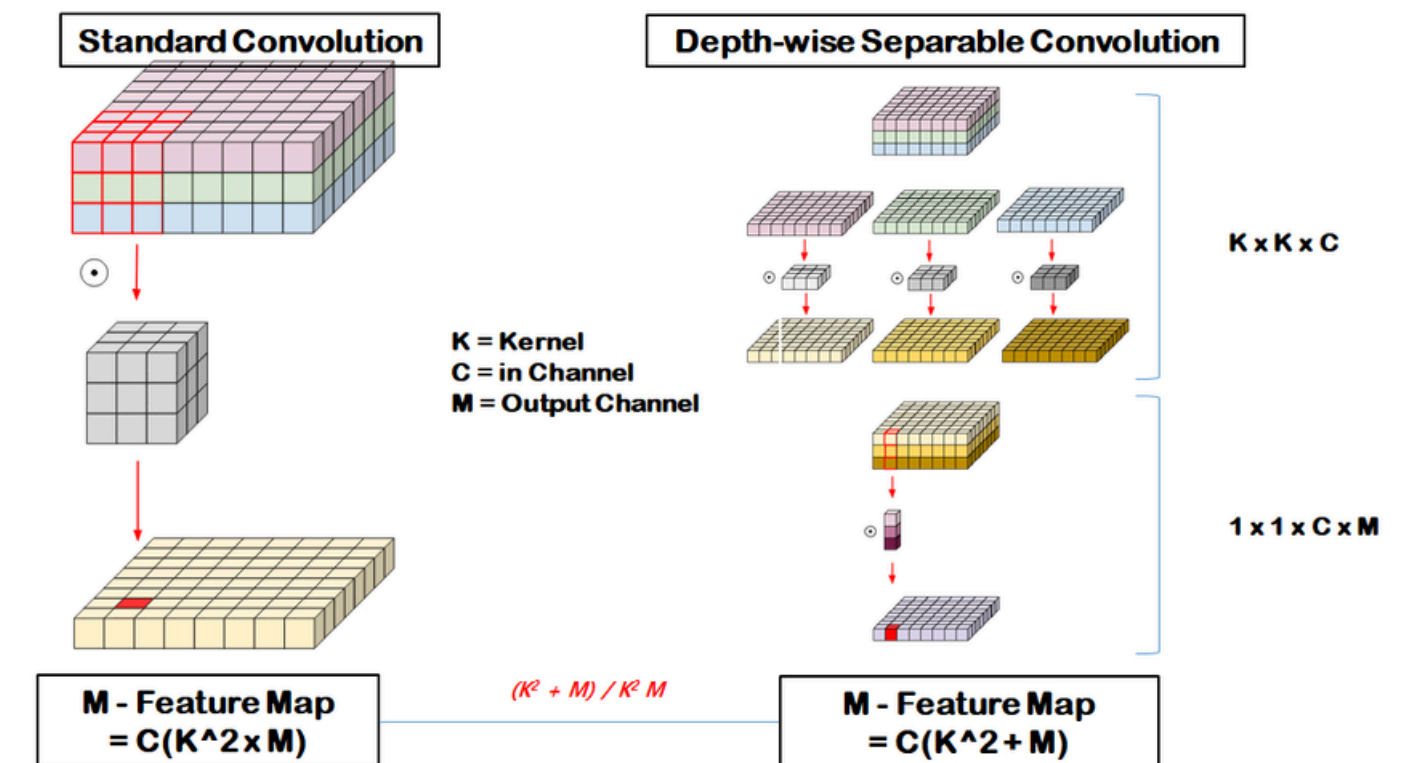


Super-resolution Image(1824*1090)





Depth-wise separable



Densenet121

Number of trainable parameters: 2.8M

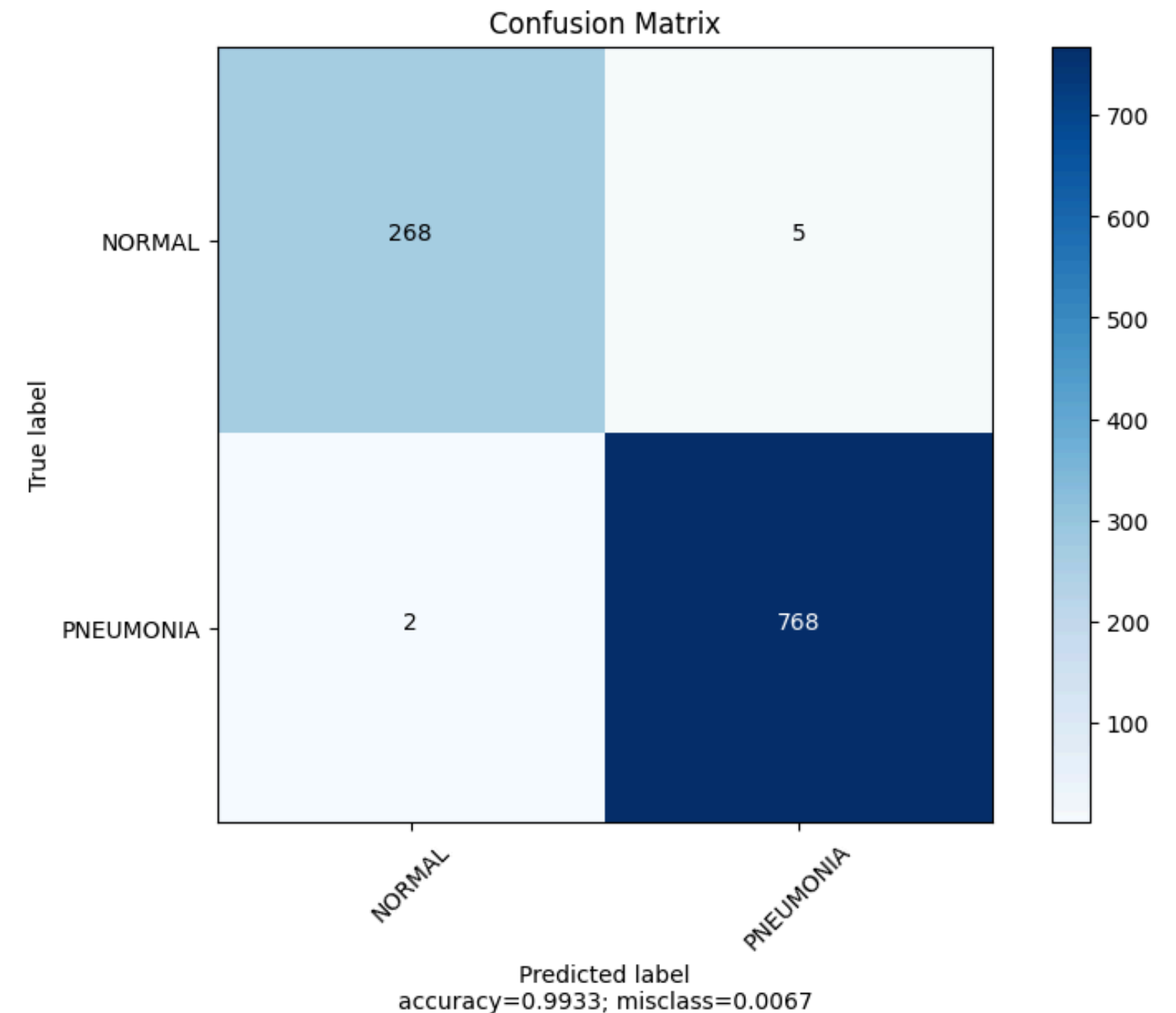
val Loss: 0.1940 | val Accuracy: 0.9223 |

Epoch 7/10

With Upscaled Image

```
e encountered in scalar divide
k = np.sum(w_mat * confusion) / np.sum(
/home/work/.local/lib/python3.10/site-pac
e encountered in scalar divide
k = np.sum(w_mat * confusion) / np.sum(
```

val Loss: 0.0380 | val Accuracy: 0.9895 |



With MovilenetV2

Number of trainable parameters: 656K(0.65M)

Epoch : 10

```
-----
Epoch 1/10
-----
0%|          | 0/522 [00:00<?, ?it/s]
train Loss: 0.2129 Acc: 0.9183
0%|          | 0/131 [00:00<?, ?it/s]
val Loss: 0.0345 Acc: 0.9904
```

```
... val Loss: 0.0292 Acc: 0.9885

Validation Classification Report:
              precision    recall  f1-score   support

   NORMAL         0.96      1.00      0.98        272
  PNEUMONIA         1.00      0.99      0.99        771

   accuracy              0.99        1043
  macro avg              0.98        1043
weighted avg              0.99        1043

Training complete in 15m 43s
Best val Acc: 0.997124
```

	ResNet	Densenet	Ours
Number of parameters	25557032	7978856	656898
Memory size(MB)	97.8	30.8	11.8

Generalization Performance

Test Classification Report:

	precision	recall	f1-score	support
NORMAL	1.0000	0.8750	0.9333	8
PNEUMONIA	0.8889	1.0000	0.9412	8
accuracy			0.9375	16
macro avg	0.9444	0.9375	0.9373	16
weighted avg	0.9444	0.9375	0.9373	16

Detailed Class Performance:

NORMAL: 87.50% (7/8)

PNEUMONIA: 100.00% (8/8)